Software Requirements Specification

For

RAJBUS-Online Bus Reservation System

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1. Introduction

1.1 Purpose

This document describes the software requirements of an online bus booking system for public transportation. It is intended for designer, developer and maintainer of the bus booking system. These requirements were created in response to the problems associated with booking bus tickets via cash.

1.2 Document Conventions

The Online Bus Reservation System is a web-based application that allows visitors check bus ticket availability, buy bus ticket and pay the bus ticket online . This system is established for all the home/office users after gaining access from the administrator. According to Invaderzim (2011), Online Bus Reservation System provides bus transportation system, a facility to reserved seats, cancellation of seats and different types of inquiry which need an instant and quick reservation. This system can be used by the users in performing online reservation via internet for their all business purposes. Users can use this program directly on their websites and no need to install it.

1.3 Intended Audience and Reading Suggestions

The use of bus traveling is a large growing business in Punjab and other states; hence bus reservation system deals with maintenance of records of each passenger who had reserved a seat for a journey. It also includes maintenance of information like schedule and details of each bus. Also, we get to know that there are many operations, which they have to do manually. It takes a lot of time and causes many errors. Due to this, sometimes a lot of problems occur and they were facing many disputes with customers. To solve the above problem, and further maintaining records of items, seat availability for customers, price of per seat, bill generation and other things, we are offering this proposal of reservation system.

1.4 Project Scope

The reservation system has three modules. First module helps the customer to enquire the availability of seats in a particular bus at particular date, the second module helps him to reserve a ticket and with the third module he can cancel a reserved ticket. The current bus booking system relies on buying tickets from the conductor for commuting to and from a location through public transportation. The task can be tedious if the number of commuters is large. Also,

payment in cash can be difficult if the payable denominations are uneven. Online Bus Booking system allows the commuter to either have a specific amount of money on his Android Based mobile, from which the ticket can be charged. Or, the costumer can buy the ticket on the bus.

2. Overview

2.1 Bus Reservation System

Bus ticket booking during the offline era posed various difficulties to the customers as well as the bus operators. Offline ticket booking reduced the scope of customers to choose different options based on their travel criterion. It also increased the franchising cost for the bus operators. At the same time, the bus operators were also finding it difficult to monitor their bus seat filling information. Many small and medium bus service organizations do not have their own online bus ticket booking system. Online Bus ticketing system web portal is a total internet ticketing operations offering the benefit of total in-house management of bus schedules, ticket bookings, ticket sales, report generation, and other business functions associated with ticket sales. It also offers the power of decision making to customers to make a ticket booking through bus operators' popularity, performance and ranking. This powerful Internet based ticket booking system that allows a full control of not only on the ticketing inventory, but also the site's content. According to Melisa, started the basic components of an Online Bus Ticketing System web portal that provides enhanced service to the bus operators and customers consist of the following:

• Capture of customer information such as name, address, phone number and e-mail address

• Price list

• Bus operators ranking

• Seating chart

• Loyalty Points/Redemption

• Search engine

• Payment information

• Organization's advertisement/slogan, phone number, fax number, and address

• Comments and suggestions section / option

• Reports

2.2 E-Ticket Reservation System

E-ticketing could be extended to major entertainment and touristic sites and thus facilitate access to major points of interest within cities, making e-ticketing also interesting for travelers. Urban tourism is the fastest growing tourism sector in the world. In public transport, e-ticketing systems are not only means of payment but process huge amount of information which offer a large range of possibilities to make public transport easier to use, to manage and to control. They offer as well opportunities to introduce integrated pricing structure that are not easy to implement with traditional payment tools. Electronic ticketing technologies are classified according to the way they are used for payment. The closer the card is to the payment system, the more reliable the transaction is, but the more constraining it is for the user . Therefore, the long-term objective is for the customer to be able to pay for public transport without having to show or validate any card, relying on fully automatic fare payment.

Public transport operators have been trying to replace paper-based tickets with electronic media, and many countries have implemented or are about to introduce e-ticketing systems. The main characteristic of eticketing is that tickets are sold and stored in electronic devices. However, the benefits of a comprehensive eticketing system for public transport operators are hard to quantify, as the main aim of e-ticketing is an improved service quality. In monetary terms, e-ticketing could reduce administrative costs as fewer cashiers are needed, fare processing times could be reduced and a better throughput of passengers could be allowed.

Moreover, fare evasion and fraud resulting from cash handling could be reduced and better price differentiation would be possible. E-ticketing enables a better integration of alternative services into the scheme, making it more attractive for customers to use it. Due to accurate data on passenger flows it might also help to better exploit the network's capacities and to improve the user experience by setting up tailor-made services for individual passengers. Costs apply that can be easily quantified, e.g., investment and operation costs, particularly the initial one-off costs (e.g., readers, software and consultancy on the scheme design). Integrated schemes appear to be particularly cost intensive, as different applications need to be connected.

Additionally, running costs for marketing, maintenance and replacement need to be considered. Costs apply for training staff or resolving passenger disputes and for setting up a (regional or even national) clearing house responsible for centralized data and fare collection. The fear of outsourcing their expertise and responsibilities in ticketing to a third party of suppliers remains a worry to public transport operators.

3. Overall Description

3.1 Product

The product is very simple in design and to implement. The system requires very low system resources and the system will work in almost all configurations. It has got following features:

• It will ensure data accuracy

• Records will be efficiently maintained by DBMS.

• Availability of seats can be enquired easily

• Passengers can also cancel their tickets easily

• Minimum time needed for the various processing

• It will provide better Service.

Shuchi, G. (2008) started that system design is to create a technical solution that satisfies the functional requirements for the system. At this point in the project life cycle there should be a Functional Specification, written primarily in business terminology, containing a complete description of the operational needs of the various organizational entities that will use the new system. The challenge is to translate all of this information into Technical Specifications that accurately describe the design of the system, and that can be used as input to System Construction.

It does not work independently. It works together with the server and users network.

Communication interface: The website communicate with the server via a communication network.

Software interface: The messages sent via the communication network are specific to the target server or user. At present, two known users will participate in the network.

Hardware interface: The software will run on an PC, Mobile, Laptop, Tablets or any digital devices yet to be chosen.

User interfaces

Customer: The customer user interface should be intuitive, such that 99.9% of all new users are able to complete their banking ticket without any assistance.

Security: It secures user personal data, their user id, password etc. It gives secure transaction when users use their ATM card or any transaction card. It is totally secure by OTP for transaction..

Maintainer: The maintainer is responsible for adding new features to the network and servicing existing product. A maintainer should be possible to add a new things to the network within 1 hour.

3.2 Product Features

The website should work 24 hrs. The website identifies a customer by a user name and password. It collects information about a simple account which is created by users. It communicates the transaction information to the customer's bank, and dispenses cash to the customer when users book their ticket. The banks allow for transaction and company gives their own software for their own computers. The software requires appropriate record keeping and security provisions. The software must handle concurrent accesses to the same account correctly.

4. Specific Requirements

4.1 System Requirements

The requirement definition is concerned with the analysis of the existing system with the aim of determining and structuring the requirement of the proposed system. It is achieved with the aid of user requirement. The Analysis stage was specifically carried out in focus of the functionality dataflow at Young Legacy Line Transport Division.

4.2 Requirement Specification

Requirement Specification a complete description of the behavior of a system to be developed and may include a set of use cases that describe interactions the users will have with the software. In addition it also contains nonfunctional requirements. Non-functional requirements impose constraints on the design or implementation (such as performance engineering requirements, quality standards, or design constraints)

4.3 Functional Requirements

Functional requirements define the specific functions that the system performs, along with the data operated on by the functions. The functional requirements are presented in scenarios that depict an operational system from the perspective of its end users. Included are one or more examples of all system features and an enumeration of all the specific requirements associated with these features:

• The system shall incorporate mechanism to authenticate its users

• The system shall verify and validate all user input and should notify in case of error detection and should help the user in error correction

• The system shall allow sharing of files in the system

• The system shall allow quick messages to be exchanged without face to face interaction

4.4 Non-Functional Requirements

Non-functional requirements address aspects of the system other than the specific functions it performs. These aspects include system performance, costs, and such general system characteristics as reliability, security, and portability. The non-functional requirements also address aspects of the system development process and operational personnel. It includes the following:

• The system shall be user friendly and consistent

• The system shall provide attractive graphical interface for the user

• The system shall allow developer access to installed environment

• The system shall target customer base

4.5 Other Requirements

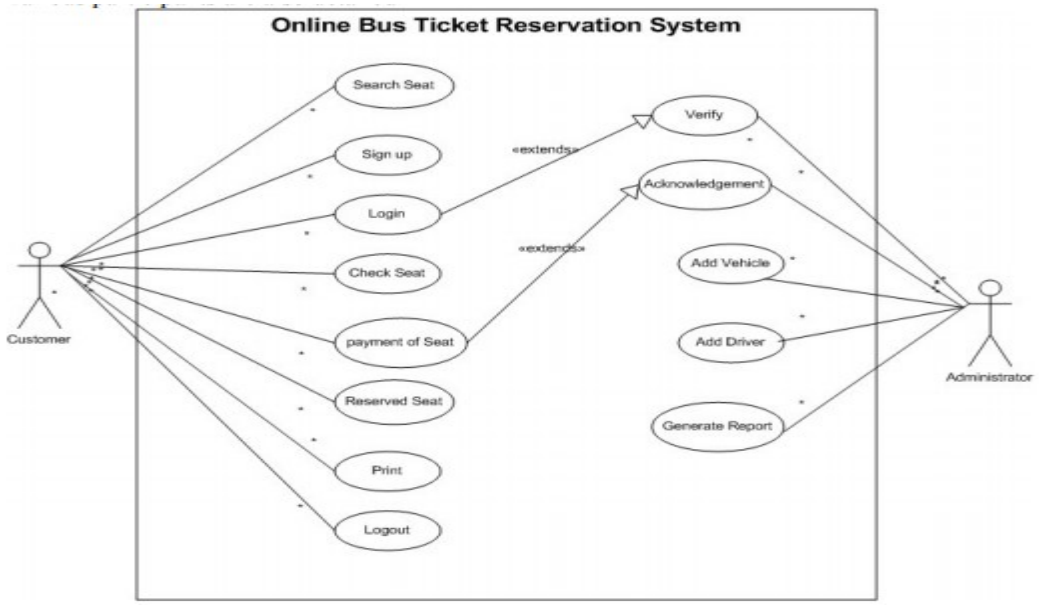
Data Base: The system must be able to use several data formats according to the data formats that are provided by the data bases of different users. The all information should have in the database.

4.6 Architecture of The Proposed System

This process supports existing infrastructure requirements and provides specific recommendations for hardware and network solutions based on existing and projected user needs. Application requirements, data resources, and people within an organization are all important in determining the optimum hardware solution. It is represented using a three tier architecture that comprises of user interface, process management and Database Management System (DBMS). It shows the components of the system, the services they provide and the way they communicate to bring about the system functionality

4.7 The Functional Model of the System

UML use case diagram of the bus reservation system is shown in the diagram below. In this figure, details of the various participants are also detailed.



4.8 User Activities

The most common activities carried out by user are illustrated bellow

• The user can search for the seat

• The user can sign up/do registration with the system

• The registered user can login to the proposed system

• The user can check for the available seat

• The user can also do payment for the seat on the proposed system

• The user can print receipt on the system as evidence of payment

4.9 Administrator Activities

The most common activities carried out by administrator are illustrated bellow

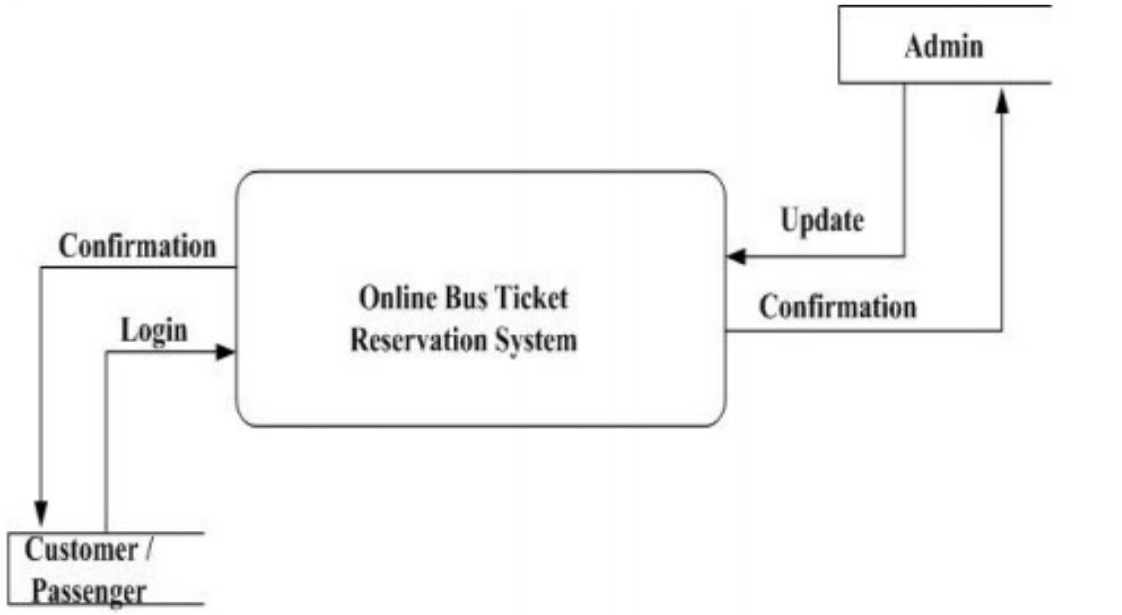
•The administrator will verify all the registered user, and allow them to login to the system

• The administrator give acknowledge to any payment user made on the system

• The administrator can add vehicle, driver and generate report as well

4.10 Data Flow Diagram

Data flow diagram is used to show the flow of data from external entities into the system. It is used to represent the physical and logical area of an information system. The data flow diagrams are pictorial or graphical representation of the Online Bus Reservation System. The data flow diagram covers all the processes and data storage area, which takes place during any transaction in the system.



5. RESULTS AND DISCUSSIONS

The proposed bus reservation system was developed using Extensible Hypertext Markup Language (XHTML), PHP Hypertext Preprocessor (PHP), Structure Query Language (SQL), Ajax, Cascading Style Sheet (CSS), and JavaScript. The relational database was adopted because is made up of a group of logically connected tables (data that has a relationship to other data). Therefore, establishing a relational database management system is a great way to increase data integrity, efficiency, ask questions, sort and filter data, provide stronger security, and share information, ease of use, data independent among others. Most of the commands used to implement processes and procedures are available in the Appendix. Some user and database table are shown in this section.

Table 1. Showing the Application Table

Server: localhost ▸ Database: transit

Structure | SQL | Search | Query | Export | Import | Operations | Privileges | Drop

	Table	Action	Records [1]	Type	Collation	Size	Overhead
☐	admin		2	MyISAM	latin1_swedish_ci	2.1 KiB	-
☐	contact		0	MyISAM	latin1_swedish_ci	1.0 KiB	-
☐	departed		1	InnoDB	latin1_swedish_ci	16.0 KiB	-
☐	driver		11	MyISAM	latin1_swedish_ci	3.0 KiB	-
☐	movement		2	InnoDB	latin1_swedish_ci	16.0 KiB	-
☐	states		10	MyISAM	latin1_swedish_ci	2.3 KiB	-
☐	teller		6	MyISAM	latin1_swedish_ci	2.3 KiB	40 B
☐	time		5	MyISAM	latin1_swedish_ci	1.1 KiB	-
☐	transit		12	MyISAM	latin1_swedish_ci	2.6 KiB	-
☐	user		9	MyISAM	latin1_swedish_ci	3.1 KiB	44 B
☐	vehicle		10	MyISAM	latin1_swedish_ci	2.4 KiB	64 B
	11 table(s)	Sum	68	MyISAM	latin1_swedish_ci	51.9 KiB	148 B

Check All / Uncheck All / Check tables having overhead With selected: ▾

Table 2. Showing the User Table

Server: localhost ▸ Database: transit ▸ Table: user

Browse | Structure | SQL | Search | Insert | Export | Import | Operations | Empty | Drop

	Field	Type	Collation	Attributes	Null	Default	Extra	Action
☐	s/n	int(11)			No	None	auto_increment	
☐	fname	varchar(30)	latin1_swedish_ci		No	None		
☐	lname	varchar(30)	latin1_swedish_ci		No	None		
☐	oname	varchar(15)	latin1_swedish_ci		No	None		
☐	title	varchar(5)	latin1_swedish_ci		No	None		
☐	address	text	latin1_swedish_ci		No	None		
☐	state	varchar(30)	latin1_swedish_ci		No	None		
☐	nation	varchar(20)	latin1_swedish_ci		No	None		
☐	phone	varchar(11)	latin1_swedish_ci		No	None		
☐	email	varchar(40)	latin1_swedish_ci		No	None		
☐	kinname	varchar(50)	latin1_swedish_ci		No	None		
☐	kinphone	varchar(11)	latin1_swedish_ci		No	None		
☐	password	varchar(15)	latin1_swedish_ci		No	None		

Check All / Uncheck All With selected:

Table 3. Showing the Admin Table

Server: localhost ▸ Database: transit ▸ Table: admin

Browse | Structure | SQL | Search | Insert | Export | Import | Operations | Empty | Drop

	Field	Type	Collation	Attributes	Null	Default	Extra	Action
☐	id	int(11)			No	None	auto_increment	
☐	username	varchar(30)	latin1_swedish_ci		No	None		
☐	password	varchar(30)	latin1_swedish_ci		No	None		
☐	phone	varchar(11)	latin1_swedish_ci		No	None		
☐	usertype	varchar(30)	latin1_swedish_ci		No	None		
☐	date	date			No	None		
☐	time	time			No	None		

Check All / Uncheck All With selected:

6. Conclusion

Nowadays, bus agencies are taking important role in transportation, and to make reservation reliable they need a strong system that they will make reservation easier, faster and safer. This project designed to meet requirements of a bus reservation system. It has been developed in XHTML, PHP, CSS, JAVASCRIPT and database has been built in MySQL. By using this application, the company can provide reservation services and information to their customers without the limitation of office hours or manpower. Not only does it let customers book trips around the clock from any location with an internet connection but it is also designed for use by the company to internally manage their business processes; minimizing human errors and overcoming difficulties and problems that arose in the previous system.